REPORT

A stereo camera is a vision system that uses two cameras to capture the same scene from slightly diff positions. It works on a principle similar to human vision, where our two eyes observe an object from different angles and helps us understand the distance and depth.

Stereo camera are commonly used in autonomous vehicles, drones and robots. In Mars rover, they help understand the surrounding terrain, estimate distance of objects and identify obstacles that may affect rover movement.

Unlike normal camera, which captures a 2D img, a stereo camera can also estimate the depth of objects and create three dimensional img.

Working Principle

↳ has 2 cameras placed at fixed distance apart.
- → Left: captures left side
- → right: captures right side.

Since, both cameras are placed at diff places, a obj can appear at slightly diff locations in the two imgs. (disparity)

Parameters

Baseline → Distance b/w two cameras
Focal length → A camera parameter that determines its img geometry
Disparity → Difference in position of same point in both imgs.

Depth Measurement

Yes, a stereo camera can estimate actual distance of an object in real-world units, such as meters or cm.
It does this using a method called triangulation. The system compares the position of same object in both imgs.

$$Z = \frac{fB}{d}$$

Object detection

(1) identification by estimating their position & height relative to surrounding.

Roads →

- Img capture: Both cameras capture img of area
- Img matching: System identifies corresponding points in both img
- Disparity calc: Calculates the difference in position of points
- Depth estimation: disparity values are converted into distance
- 3D mapping: Depth information is used to generate a 3D point cloud of the surrounding.
- Obstacle detection: The system analyses the point ~~cloud~~ cloud such as rocks, boulder & other obstacles.
- Navigation: Rover determines whether it can safely cross the obstacles or needs to change its route.

Ditch / Hole detection

↳ Yes, stereo camera can detect them, they are called negative obstacles.

This prevents the rover in getting stuck in a ditch or hole & damage itself.

The detection works by analysis of ground depth & height.

Process →

- The stereo camera capture imgs of the terrain in front.
- The system generate a depth map or 3D point cloud.
- It estimates the normal ground surface using surrounding terrain
- If there is a sudden drop in ground or a gap in depth information, the system identifies a possible depression
- The system evaluates its size, depth and position to determine whether it is safe to cross
- If depression is considered dangerous, the rover avoids that region & plan another route.

Role of AI

Stereo vision provides information abt distance & position of objects while AI helps rover understand what these objects are & how they may affect its movement.

Object detection: AI models can identify rocks, boulders, rovers
Terrain classification: distinguish b/w diff surfaces such as sand, gravel, rocks.
obstacle recognition: By combing image information with depth measurement, identify potential obstacles
path planning: After identifying unsafe areas, navigation system select a suitable path toward its destination

This allows rover to make basic navigation decisions w/ requiring human instructions for every movement.

Application in Mars Rover

↳ Stereo camera are particularly useful in planetary exploration bcs communication b/w Earth & Mars involves delays. Therefore rovers needs their surroundings and make navigation decision independely. (eg - Mazcam, NAVCams)

↳ Detecting nearby rocks & other obstacles
↳ Identifying potentially dangerous terrain
↳ Creating 3D representation of surrounding
↳ Supporting autonomous navigation

Stereo Camera VS LIDAR

Stereo

- uses img & triangulation
- Ditch detection possible
- Available w/ color camera
- limitation - matching error, poor lighting

LIDAR

- Uses laser to measure distance
- Ditch detection NOT
- in conventional LiDAR
- Limitation - Range & Surface reflectivity

Date
Page

Limitations

- Poor Lighting: Low light can make it difficult to match points
- Smooth Surface: Surfaces w/ very little texture, such as uniform sand can make depth estimation difficult
- Reflective surfaces: Reflections can lead to incorrect depth measurements
- Long distances: Depth measurements become less accurate.
- Obstructive views: If an object is visible to only one camera, accurate depth estimation may not be possible.

To improve reliability, Stereo cameras can be used along with other sensors as LiDAR, IMUs & wheel odometry.